



NEBULA SYNDICATE
HEADQUARTERS ASTRA IMPERIALIS TERRANUM



4 May 2026

MEMORANDUM FOR COUNT VIKTOR THUNDERCLAW

FROM: Nebula Syndicate Network (NSN) Administratum

SUBJECT: Meeting Minutes – NSN Security and Recommendations

1. Network administrators have expressed concerns about the security posture of the NSN. This meeting aired administrator and security grievances. Ultimately, the final say is yours, Count Victor Thunderclaw.

2. Internet Connectivity:

- a. While the NSN's connection to the internet is protected by a top-of-the-line Small Office/Home Office (SOHO) router with a firewall capability, no one is maintaining it or monitoring the logs or alerts – partly due to the last administrator defecting to the Aurora Alliance (AA). Understaffed Administratum Security do have the capacity to pick-up this duty.
- b. The exact posture of the firewall rules is unknown. A fear exists that some rules may leave large parts of the internal NSN accessible from the public Internet.

3. Lack of Internal Network Segmentation and Security:

- a. The Development Network, used to develop the software for the Space Laser, is directly accessible to the Internet through the SOHO router – with no additional network security or segmentation.
- b. That same SOHO router, in a flat topology, connects the Administrative/Evil Business Network, Development/Test Network, and the Mission Operations Center that manages the Nebula Syndicate Ground Station Network (NSGSN). The administrative level and mission control components connected to the Internet need robust firewalling and intrusion detection systems to mitigate these risks.

4. Data Encryption:

- a. AA persistent foiling of our financial investments has dampened our attempts to upgrade/update legacy systems. There are several parts of the network that do not use secure protocols or continue to use legacy protocols to ensure older systems continue to

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function as intended. For example, TELNET is heavily used throughout the network specifically in the Nebula Syndicate Ground Station Network (NSGSN).

5. Mission Operations Center (MOC):

- a. The Mission Control Systems are critical and should be protected with strict access controls, rigorous authentication processes, and physical security measures to prevent unauthorized access. We continue to find syndicate personnel sharing user accounts and weak passwords. Despite our strict policy and procedures, personnel continue to be our weakest link.
- b. On similar note, there is causality between the current “corrective action” for not abiding to policy and procedure and our manning problems; please consider not purging offenders on the first offence?

6. MOC and Ground Station Physical Security:

- a. All the funds allocated for physical security at the MOC and the four ground station locations were diverted to the development of the Space Laser. The locations use legacy RFID door locks to secure against tampering, sabotage, and unauthorized physical access. The AA has prevented us on numerous occasions from acquiring more “state of the art” technology, yet we feel the current technology continues to work as intended; we have not had any physical security breach attempts recorded in the logs.

7. Network Monitoring:

- a. The MOC and the rest of the NSN lacks sufficient continuous monitoring of the network for unusual activities or anomalies. NSN Administratum Security is unable to detect and respond to potential security threats promptly.
- b. On similar note, we need to assimilate new developers to develop the security solution. We had to take the last security solution offline due to AA Cyber Command supply chain attack which compromised one of the NSGSN ground sites. Unfortunately, the last Director of NSN Administratum, Overseer Netscream, was investigating the matter, but failed you for the last time and well... they’re no longer with us.

6. The Space Laser Communication Links:

- a. It is essential to implement encryption measures to safeguard radio frequency (RF) links between the ground station antenna and the Space Laser Satellite against potential eavesdropping; however, these currently operate without encryption. This means that the bulk transmissions from the ground station to the satellite are unencrypted. While the traffic within the RF signal could use encryption, this is also lacking.
- b. There is no authentication mechanism in place for messages sent to the Space Laser Satellite. This oversight leaves the system vulnerable to replay attacks, where a

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sophisticated attacker could capture and retransmit communications to compromise satellite operations.

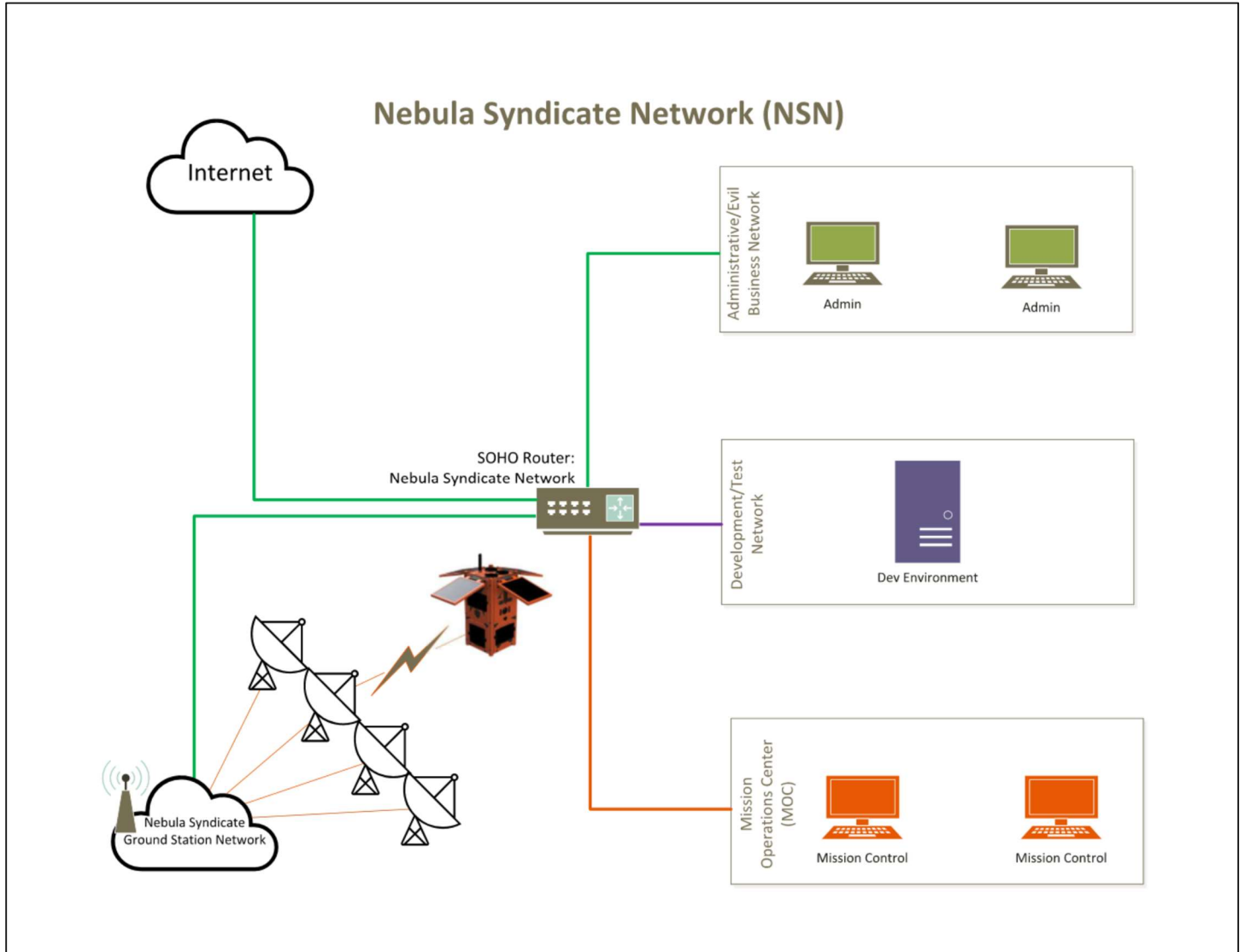
- c. The absence of command sequence tracking in the system's operations is a significant concern. Implementing command sequence tracking would alert satellite operators to any anomalies in the sequence numbers, such as out of order commands, which could indicate malicious interference with the satellite.
- d. The Consultative Committee for Space Data Systems (CCSDS) Space Packet Protocol, which is currently used for the link, supports an extension known as the Space Data Link Security Protocol. This extension could be effectively utilized to enhance the security of the traffic between the ground station and the satellite, thereby mitigating the identified vulnerabilities. This protocol can both encrypt and authenticate the space packets.

//SIGNED//
OVERSEER PACKETSTORM
Director, NSN Administratum

Attachments:

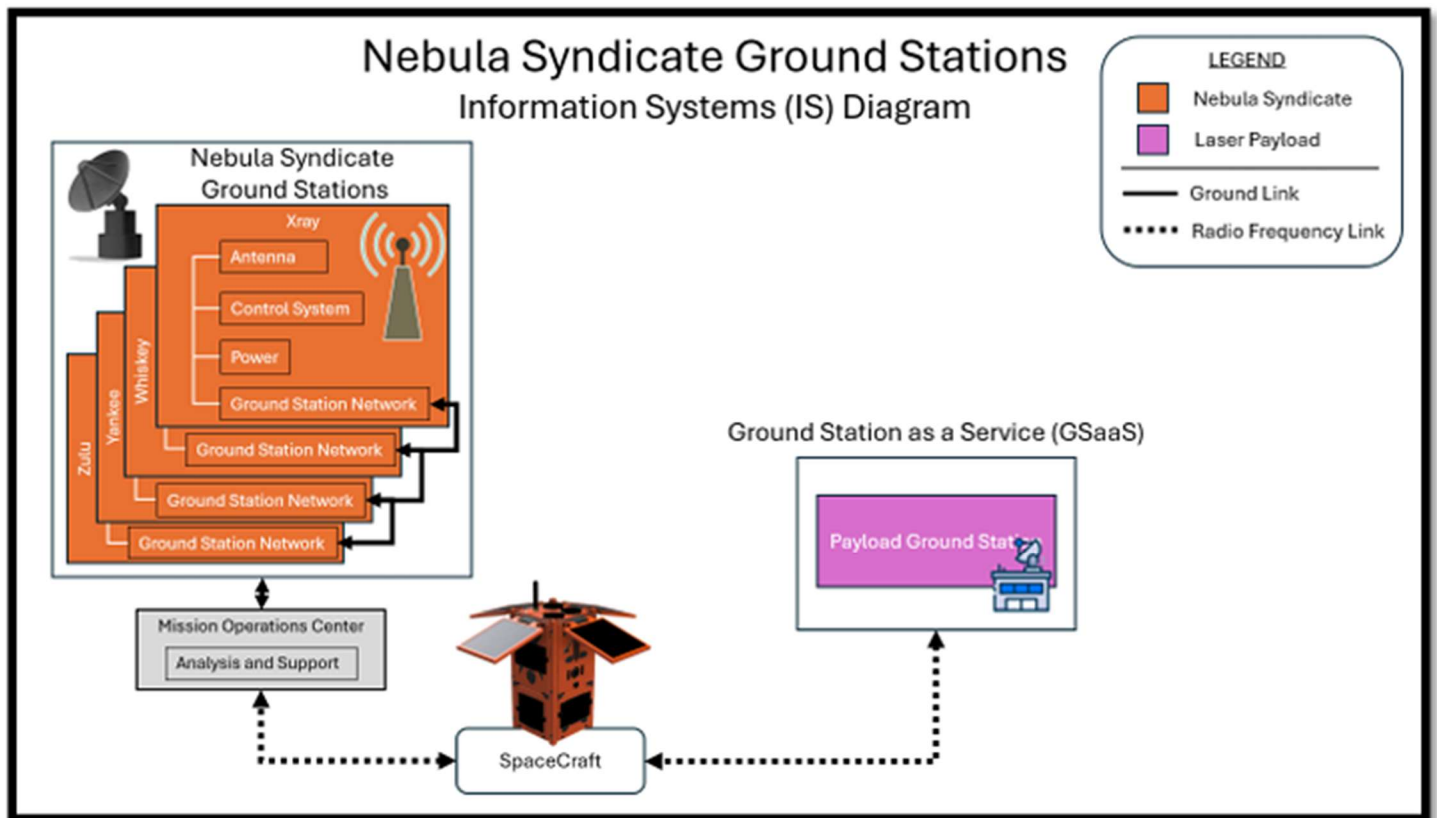
- 1. NSN Topology
- 2. Nebula Syndicate Ground Station Architecture

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Nebula Syndicate Space Laser Ground Station Architecture

Current as of: June 2024



The Nebula Syndicate Information Systems (IS) Diagram for Ground Stations presented here provides a comprehensive overview of the architecture and connectivity involved in satellite ground operations. The Nebula Syndicate owns and operates its own ground stations but outsources the commanding of its Space Laser through a Ground Station as a Service (GSaaS):

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Nebula Syndicate Ground Stations:

- **Antenna:** The primary physical interface for communication with spacecraft, responsible for both receiving signals from and transmitting signals to the spacecraft. Four (4) Nebula Syndicate Ground Stations exist around the globe. The spacecraft experiences times of coverage from ground antennas and then periods between contacts.
- **Control System:** Manages the operations of the antenna and the overall ground station, including signal processing and command dispatch.
- **Power:** Essential infrastructure that supplies and manages the power requirements for the ground station systems.
- **Ground Station Network:** A series of interconnected systems that ensure robust communication channels within the ground station and with other networks.
- **WIFI Access:** Each ground station location employs a WIFI network with WPA2 encryption. This network is used to connect IOT devices and other remote devices that help the antennas function, move, and track Space Laser Satellite as it moves across the sky.
 - The previous Director, Overseer Netscream, created a very strong passphrase aka password for the WIFI access point and has not been updated in years and is used at each ground station location. The IT security team has recommended it be rotated soon. Provided is the encoded version, for security purposes:
Q1QzezdoQHQnc19hbWF6MW5nIV9JX2hAdjNfdGgzX3NAbTNfYzBtYmluQHRpb25fdDBfbXlfbHVnZ0BnZSF9

Mission Operations Center (MOC):

- Located downstream of the Nebula Syndicate Ground Stations, the MOC handles the comprehensive management of the mission, including data analysis and operational support tasks. It serves as the hub for overseeing the spacecraft and payload operations, ensuring that mission objectives are met.

Ground Station as a Service (GSaaS) for the Space Laser:

- This is a service model where ground station capabilities are offered as a service to payload owners, enabling them to communicate with their space assets without the need to own and operate ground station infrastructure. (Note: Several real world GSaaS exist such as capabilities from AWS Ground Station and Microsoft Azure Orbital.)
- **Space Laser:** Because GSaaS often has worldwide coverage, commands to fire the Space Laser can be executed at any time.
- Security of GSaaS is often a shared responsibility between the customer and service provider.

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